

# Luca Lazzaroni

**Assistant Professor in Electronics**

DITEN – UNIGE

BASED IN Genoa, Italy

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SECTOR RTD-A · IINF-01/A

BIBLIOMETRICS — SCOPUS

| 68 | 345 | 11 | 21 |
|---|---|---|---|
| DOCUMENTS | CITATIONS | H-INDEX | ARTICLES, 5 YRS |

SYNCED OCT 26, 2026

## 01 Profile

Assistant Professor (RTD-A) at the Department of Naval, Electrical, Electronic and Telecommunications Engineering (DITEN), University of Genoa, sector IINF-01/A – Electronics, since March 2025. PhD in Science and Technology for Electronic and Telecommunication Engineering since 2024.

Research on embedded machine learning and edge AI: TinyML toolchains, quantized and binary neural networks on microcontrollers and FPGAs, perception and decision-making for automated driving, deep reinforcement learning, on-device LLMs. The work is also carried out within European and national research projects with industrial partners.

Edge AI · TinyML · Embedded deep learning · Automated driving · Deep reinforcement learning · On-device LLMs

## 02 Contact & profiles

| | | |
|---|---|---|
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| LINKEDIN luca-lazzaroni | | |

## 03 Appointments & education

### POSITIONS

Mar 2025 – present

**Assistant Professor (RTD-A)**

DITEN, University of Genoa

Scientific sector IINF-01/A — Electronics.

| | |
|---|---|
| Oct 2023 — Feb 2025 | **Research Fellow**<br>DITEN, University of Genoa<br>Grant: “Machine Learning models for the processing of data from autonomous vehicles”. |
| Nov 2020 — Oct 2023 | **PhD Candidate**<br>DITEN, University of Genoa — XXXVI cycle<br>Scholarship: “Deep learning on the edge”. |

### EDUCATION

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|---|---|
| 2024 | **PhD in Science and Technology for Electronic and Telecommunication Engineering**<br>University of Genoa — awarded March 14, 2024<br>Thesis: “The Facets of Edge AI in Automotive: Exploring Embedded Frameworks, Voice Assistants, and Deep Reinforcement Learning” |
| 2020 | **MSc in Electronic Engineering (LM-29)**<br>University of Genoa — July 24, 2020<br>Thesis: “Design and implementation of a multiplatform execution engine for IoT applications” · Grade: 110/110 |
| 2017 | **BSc in Electronic Engineering and Information Technologies (L-8)**<br>University of Genoa — October 27, 2017<br>Thesis: “Model-based design of Lego robot control in Simulink environment” |
| 2014 | **Scientific High School Diploma**<br>Liceo Scientifico Statale “Arturo Issel”, Finale Ligure (SV)<br>Grade: 100/100 |

Licensed Information Engineer — State Examination passed in the second session of 2020.

## 04 Teaching & academic appointments

### COURSES

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|---|---|
| A.Y. 2024/25 – 2026/27 | **Distributed Electronic Systems and Technologies for Environmental Monitoring**<br>Course lecturer · cod. 94631 |
| A.Y. 2024/25 – 2026/27 | **Tiny Machine Learning**<br>Co-lecturer · cod. 114752 |
| A.Y. 2026/27 | **Electronic Devices and Circuits**<br>Course lecturer · cod. 65997 |
| since A.Y. 2022/23 | **Machine Learning for Automated Driving**<br>Subject expert · cod. 108716 |
| A.Y. 2023/24 | **Computer Graphics Arch. — Games and Simulation**<br>Adjunct professor · cod. 172317 |

| | |
|---|---|
| A.Y. 2022/23 | **Autonomous Agents in Games**<br>Teaching support · cod. 198216 |
| A.Y. 2020/21 | **Fundamentals of Programming of Electronic Systems**<br>Teaching support · cod. 166061 |
| A.Y. 2020/21 | **Makers Approach to Electronic Design**<br>Teaching support · cod. 186985 |

All courses at the University of Genoa.

### SUPERVISION

**1** PhD candidate — Ali Zare, Science and Technology for Electronic and Telecommunication Engineering (XLI cycle)

**12** BSc theses — Electronic Engineering and Information Technologies (L-8)

**10** MSc theses — Electronic Engineering (LM-29)

**7** MSc theses — Technological Engineering for Strategy and Security (LM/DS)

**2** MSc theses — Computer Science (LM-18)

### ACADEMIC APPOINTMENTS

**Member of the Teaching and Admissions Committee, MSc in Engineering for Natural Risk Management.**

**Communications officer for the MSc in Electronic Engineering.**

## 05 Editorial & service

### EDITORIAL ROLES

2026 — **Volume Editor**
APPLEPIES 2026 — Springer LNEE
Proceedings in preparation.

2025 — **Editorial Board Member (Early Career)**
Sensors — MDPI
Since December 18, 2025.

2025 — **Volume Editor**
APPLEPIES 2025 — Springer LNEE
Proceedings of the International Conference on Applications in Electronics Pervading Industry, Environment and Society.

2024 — **Guest Editor**
Energies — MDPI
Special Issue “Tiny Machine Learning for Energy Applications”, April 2024 — August 2025.

## CONFERENCE ORGANIZATION

**Program Chair**
APPLEPIES 2026
Bologna, Italy, September 10–11, 2026.

**Program Chair**
APPLEPIES 2025
Turin, Italy, September 11–12, 2025.

**Program Committee**
APPLEPIES
Since 2023.

**Program Committee**
GALA — Games and Learning Alliance
Since 2023.

**Session Chair**
APPLEPIES 2023
1st Short Oral session — Genoa, Italy, September 28–29, 2023.

## PEER REVIEW

### JOURNALS

| | |
|---|---|
| IEEE Access | since 2023 |
| Journal of Systems Architecture | since 2023 |
| Engineering Applications of Artificial Intelligence | since 2024 |
| Sensors | since 2024 |
| IET Intelligent Transport Systems | since 2025 |
| IEEE Transactions on Intelligent Vehicles | since 2026 |
| IEEE Transactions on Circuits and Systems II: Express Briefs | since 2026 |
| Robotics and Autonomous Systems | since 2026 |
| Scientific Reports | since 2026 |
| Journal of Big Data | since 2026 |
| Discover Internet of Things | since 2026 |

### CONFERENCES

| | |
|---|---|
| IEEE ICRA — International Conference on Robotics and Automation | since 2026 |
| IEEE/RSJ IROS — Intelligent Robots and Systems | since 2025 |
| GALA — Games and Learning Alliance | since 2023 |
| APPLEPIES | since 2021 |

## 06 Invited & conference talks

13 TALKS

| | |
|---|---|
| Sep 11–12, 2025 | **APPLEPIES 2025**<br>Turin, Italy |
| Jan 11–14, 2025 | **43rd IEEE ICCE — International Conference on Consumer Electronics**<br>Las Vegas, NV, USA |
| Sep 19–20, 2024 | **APPLEPIES 2024**<br>Turin, Italy |
| Jun 24–26, 2024 | **55th Annual Meeting of the Italian Society of Electronics (SIE)**<br>Genoa, Italy |
| Sep 28–29, 2023 | **APPLEPIES 2023**<br>Genoa, Italy |
| Sep 7–9, 2023 | **54th Annual Meeting of the Italian Society of Electronics (SIE)**<br>Noto (RG), Italy |
| Sep 26–27, 2022 | **APPLEPIES 2022**<br>Genoa, Italy |
| Sep 7–9, 2022 | **53rd Annual Meeting of the Italian Society of Electronics (SIE)**<br>Pizzo (VV), Italy |
| Sep 21–22, 2021 | **APPLEPIES 2021**<br>Pisa, Italy |
| Dec 1–2, 2021 | **GALA 2021 — Games and Learning Alliance**<br>La Spezia, Italy (online) |
| Jul 7–9, 2021 | **52nd Annual Meeting of the Italian Society of Electronics (SIE)**<br>Trieste, Italy (online) |
| Dec 9–10, 2020 | **GALA 2020 — Games and Learning Alliance**<br>Laval, France (online) |
| Nov 19–20, 2020 | **APPLEPIES 2020**<br>Online |

## 07 Awards & memberships

2025

**Best Presentation of the Session — “Edge Computing”**

43rd IEEE International Conference on Consumer Electronics (ICCE), Las Vegas, NV, USA, January 11–14, 2025

A. Dabbous, L. Lazzaroni, F. Sakr, R. Berta, M. Fresta, F. Bellotti — “Binary Neural Networks for Real-Time Structural Health Monitoring on Edge Devices”

PROFESSIONAL MEMBERSHIPS

**IEEE**

IEEE Member and IEEE Young Professional.

**IEEE Consumer Technology Society**

Member of the “Machine Learning, Deep Learning and AI in CE (MDA)” Technical Committee.

**SIE — Italian Society of Electronics**

Member within the University of Genoa Research Unit since 2021, research area “Electronic systems and applications”.

## 08 Research projects

Synergies

**Generation of scenarios for development, training, virtual testing and validation of CCAM systems**

European — HORIZON-CL5-2023-D6-01-02 · Task Leader, WP8 — Evaluation Framework: T8.1 and T8.4 (Future Planning)

Generates test scenarios for training, testing and validating CCAM systems, focusing on urban and rural traffic — where relevant-scenario knowledge is far thinner than for highway driving. Runs with most major European carmakers and universities active in the field.

Hi-Drive

**Addressing challenges toward the deployment of higher automation**

European — H2020, GA 101006664 · Member of the University of Genoa research unit

Develops and tests advanced automated driving at scale, moving from conditional automation (SAE L3) to high automation across different scenarios and road contexts.

L3Pilot

**Piloting Automated Driving on European Roads**

European — H2020 · Member of the University of Genoa research unit

Large-scale piloting of SAE L3 automated driving functions on public roads across Europe, with the major European carmakers and suppliers.

GEMINI

**Green Machine Learning for the IoT**

National — PRIN 2022 · Member of the University of Genoa research unit (DII–UNITN / DITEN–UNIGE)

A framework for fully sustainable TinyML applications deployed along an edge-to-cloud continuum, targeting battery-free embedded devices — including those with no network connectivity at all.

ADINA

**Integrated Digital Assistant Automotive**

Regional — POR FESR Piedmont 2014-2020 · Member of the University of Genoa research unit, with Lifetouch

An in-vehicle virtual assistant on dedicated hardware, connected to the car's systems: AI algorithms interpret the driver's requests and act on vehicle and infotainment functions.

## 09 Publications

74 ITEMS · 68 IN SCOPUS

2026

**Cross-platform deployment and characterization of an on-device generative conversational agent**
P. Firpo, R. Berta, F. Bellotti, **L. Lazzaroni**, M. Fresta, A. Pighetti, A. Abedinia, N. Yoonesi, D.P. Pau
*Journal of Systems Architecture* vol. 181, art. 103981 Journal

**A Deployment-Oriented Simulation Framework for Deep Learning-Based Lane Change Prediction**
L. Forneris, R. Berta, M. Fresta, **L. Lazzaroni**, H. Rojhan, C. Oh, A. Pighetti, H. Ballout, F. Tango, F. Bellotti
*IEEE Signal Processing Letters* vol. 33, pp. 136–140 Journal

**Efficient Certified Robustness Assessment for Transformer Models via Verifier-Aware Perturbation-Position Scheduling**
V. Soltanmuradov, R. Berta, **L. Lazzaroni**, D.M. Gómez, A. Pighetti, F. Bellotti
*Machine Learning and Knowledge Extraction* vol. 8, no. 9, art. 290 Journal

**FPGA Deployment of Extremely Quantized Models for Time-to-Lane-Change Prediction**
A. Saad, R. Berta, M. Fresta, **L. Lazzaroni**, A. Pighetti, O. Srour, F. Tango, F. Bellotti
*IEEE Transactions on Circuits and Systems II: Express Briefs* vol. 73, no. 8, pp. 888–892 Journal

**Usability, Technicality, Interoperability (UTI), a Comprehensive Assessment Framework for Digital Platforms**
H. Rojhan, F. Bellotti, M. Fresta, **L. Lazzaroni**, R. Berta
*IEEE Access* vol. 14, pp. 44886–44903 Journal

**StabilEdge, Formal Verification of Edge AI Applications**
**L. Lazzaroni**, R. Berta, V. Soltanmuradov, D. Martin Gomez, F. Bellotti
*IEEE Access* vol. 14, pp. 37416–37434 Journal

**Resource-Adaptive FPGA-Based Kernels for Accelerating Self-attention in Transformers**
A. Saad, F. Bellotti, H. Ballout, M. Fresta, **L. Lazzaroni**, O. Srour, R. Berta
*Applications in Electronics Pervading Industry, Environment and Society* Lecture Notes in Electrical Engineering
pp. 321–327 Conference

**Preface**
M. Ruo Roch, R. Berta, A. Gómez, **L. Lazzaroni**, M. Martina, P.M. Ros, N. Pazos Escudero
*Lecture Notes in Electrical Engineering* pp. v–vi Editorial

**Optimizing Binary AutoEncoders on STM32 Microcontrollers**
H. Ballout, R. Berta, **L. Lazzaroni**, F. Sakr, A. Pighetti, M. Fresta, F. Bellotti
*Applications in Electronics Pervading Industry, Environment and Society* Lecture Notes in Electrical Engineering
pp. 301–307 Conference

**Modeling Power Electronics in Simulink. A Tutorial for MSc Electronic Engineering Students**
L. Motta, F. Bellotti, R. Berta, **L. Lazzaroni**, M. Fresta, N. Busi
*Applications in Electronics Pervading Industry, Environment and Society* Lecture Notes in Electrical Engineering
pp. 242–248 Conference

**Managing Performance Indicator (PI) Forms Through a State-of-the-Art Big Data Platform**
H. Rojhan, F. Bellotti, **L. Lazzaroni**, M. Fresta, L. Forneris, A. Pighetti, R. Berta
*Applications in Electronics Pervading Industry, Environment and Society* Lecture Notes in Electrical Engineering
pp. 218–223 Conference

**Lightweight Deep Learning for SOC Estimation of Various Lithium-Ion Batteries on Xilinx ZCU104 FPGA**
D. Faryar, R. Berta, M. Fresta, A. Saad, **L. Lazzaroni**, H. Ballout, O. Srour, F. Bellotti
*Applications in Electronics Pervading Industry, Environment and Society* Lecture Notes in Electrical Engineering
pp. 25–31 Conference

**Improving Lane Change Performance in CARLA Through Curvature-Adaptive B-Spline Planning**
O.L. Audi, F. Bellotti, R. Berta, Z. Liu, **L. Lazzaroni**, A. Pilla, H. Rojhan, L. Forneris
*Applications in Electronics Pervading Industry, Environment and Society* Lecture Notes in Electrical Engineering
pp. 212–217 Conference

**Implementing YoloV8 on Zynq UltraScale+MPSoC FPGA Boards**
H. Ballout, R. Berta, O. Srour, **L. Lazzaroni**, M. Fresta, A. Saad, F. Bellotti
*Proceedings of SIE 2025* Lecture Notes in Electrical Engineering pp. 289–294 Conference

**Generation of Synthetic Datasets of Highway Driving Scenarios**
M. Cossu, R. Berta, **L. Lazzaroni**, L. Forneris, M. Fresta, A. Pighetti, A. Pilla, J.L. Sauvaget, H. Touil, L. Durville, T. Griffon, G.B. Nejma, F. Hadj Selem, F. Bellotti
*IEEE Transactions on Intelligent Transportation Systems* Journal

**FPGA Implementation of a Robust Decision-Making System for Low-Speed Maneuvering in Automated Driving**
A. Saad, H. Ballout, F. Bellotti, M. Fresta, **L. Lazzaroni**, O. Srour, R. Berta
*Proceedings of SIE 2025* Lecture Notes in Electrical Engineering pp. 295–301 Conference

**Exploring Neural Architecture Search (NAS) for Binary Neural Networks (BNNs) in the New WakeVision**
A. Pighetti, F. Bellotti, S. Mohammady, **L. Lazzaroni**, H. Ballout, R. Berta
*Applications in Electronics Pervading Industry, Environment and Society* Lecture Notes in Electrical Engineering
pp. 85–90 Conference

**Embedded Deployment of an LLM-Based Voice Assistant**
**L. Lazzaroni**, F. Bellotti, A. Pighetti, A. Saad, O. Srour, R. Berta
*Applications in Electronics Pervading Industry, Environment and Society* Lecture Notes in Electrical Engineering
pp. 273–278 Conference

**Comparing Complete and Incomplete Formal Verification in Deep Learning-Based Object Detection**
V. Soltanmuradov, F. Bellotti, R. Berta, D.M. Gómez, **L. Lazzaroni**
*Applications in Electronics Pervading Industry, Environment and Society* Lecture Notes in Electrical Engineering
pp. 286–291 Conference

**A Dual Branch Quantized Neural Network for Road Surface Classification Using Smartphone Sensors**
S.G. Fashkhami, R. Berta, A. Pighetti, M. Fresta, **L. Lazzaroni**, H. Ballout, H. Rojhan, F. Bellotti
*Applications in Electronics Pervading Industry, Environment and Society* Lecture Notes in Electrical Engineering
pp. 183–190 Conference

**A Collaborative GitHub Framework for Project-Driven Learning in Microcontroller-Based Embedded Systems**
E. Castagno, F. Bellotti, R. Berta, **L. Lazzaroni**, M. Fresta, S. Incerti
*Applications in Electronics Pervading Industry, Environment and Society* Lecture Notes in Electrical Engineering
pp. 249–254 Conference

**A CARLA Parking Environment for Training Deep-Reinforcement Learning Agents**
A. Zare, F. Bellotti, R. Berta, **L. Lazzaroni**, L. Forneris, A. Pighetti
*Applications in Electronics Pervading Industry, Environment and Society* Lecture Notes in Electrical Engineering
pp. 79–84 Conference

2025

**An Embedded System Architecture for Robust Deep Learning-Based Traffic Light Detection**
**L. Lazzaroni**, F. Bellotti, A. Pighetti, L. Forneris, M. Fresta, H. Ballout, C. Oh, S. Pantawane, R. Berta
*IEEE Sensors Journal* vol. 25, no. 24, pp. 44859–44874 Journal

**TinyML Acceleration with MAX78000**
A. Dabbous, **L. Lazzaroni**, F. Bellotti, S.M. Presentación, A. Pighetti, R. Berta
*Proceedings of SIE 2024* Lecture Notes in Electrical Engineering pp. 468–474 Conference

**Setting Up a Lightweight Simulation Environment for Automated Driving Dataset Collection**
L. Forneris, F. Bellotti, R. Berta, M. Cossu, M. Fresta, R. Rezaieamale, H. Rojhan, V. Soltanmuradov, F. Tango, **L. Lazzaroni**
*Applications in Electronics Pervading Industry, Environment and Society* Lecture Notes in Electrical Engineering pp. 156–163 Conference

**Runtime Input Monitoring of a Reinforcement Learning-Based Agent for Automated Car Parking**
H. Ballout, R. Berta, I. Bochenko, **L. Lazzaroni**, A. Dhonthi, V. Hashemi, F. Bellotti
*Applications in Electronics Pervading Industry, Environment and Society* Lecture Notes in Electrical Engineering pp. 387–394 Conference

**Robustness Verification of a Reinforcement Learning-Based Agent for Automated Car Parking**
F. Bellotti, R. Berta, V. Soltanmuradov, D.M. Gómez, A. Dhonthi, V. Hashemi, **L. Lazzaroni**
*Applications in Electronics Pervading Industry, Environment and Society* Lecture Notes in Electrical Engineering pp. 139–147 Conference

**RL-Based Generation of a Synthetic Automotive Driving Scenario Dataset**
R. Berta, F. Bellotti, M. Cossu, L. Forneris, **L. Lazzaroni**, A. Pighetti
*Applications in Electronics Pervading Industry, Environment and Society* Lecture Notes in Electrical Engineering pp. 395–402 Conference

**RAFT: Regularized Adversarial Fine-Tuning to Enhance Deep Reinforcement Learning for Self-Parking**
A. Pighetti, F. Bellotti, R. Berta, A. Cavallaro, **L. Lazzaroni**, C. Oh
*IEEE Sensors Letters* vol. 9, no. 9, art. 6010204 Journal

**Quantization of MobileNetV2 for Resource-Constrained Microcontrollers**
**L. Lazzaroni**, F. Bellotti, A. Dabbous, A. Pighetti, R. Berta
*Applications in Electronics Pervading Industry, Environment and Society* Lecture Notes in Electrical Engineering pp. 497–504 Conference

**Performance Comparison of YOLOv8 and YOLOv10 for Traffic Light Detection on a Jetson Nano Board**
H. Ballout, R. Berta, A. Dabbous, M. Fresta, **L. Lazzaroni**, A. Pighetti, F. Bellotti
*Applications in Electronics Pervading Industry, Environment and Society* Lecture Notes in Electrical Engineering pp. 242–249 Conference

**On-Board Deployability of a Deep Learning-Based System for Distraction and Inattention Detection**
M. Fresta, F. Bellotti, **L. Lazzaroni**, H. Ballout, A. Pighetti, Ó. Ameneiro-Prieto, M. Porta-Lorenzo, D. Sánchez-Louzán, C. González-Escudeiro, F. Tango, R. Berta
*IEEE Open Journal of the Industrial Electronics Society* vol. 6, pp. 1351–1369 Journal

**Leveraging Neural Architecture Search for Structural Health Monitoring on Resource-Constrained Devices**
A. Capello, R. Berta, M. Fresta, **L. Lazzaroni**, F. Bellotti
*Applications in Electronics Pervading Industry, Environment and Society* Lecture Notes in Electrical Engineering pp. 323–330 Conference

**Green Machine Learning for the IoT (GEMINI): A Position Paper**
**L. Lazzaroni**, A. Dabbous, M. Nardello, M.A. Lebdeh, R. Berta, K.S. Yildirim, F. Bellotti, D. Brunelli
*Applications in Electronics Pervading Industry, Environment and Society* Lecture Notes in Electrical Engineering
pp. 339–346 Conference

**Exploring Decision Transformer for Highway Automated Driving**
L. Forneris, F. Bellotti, R. Berta, **L. Lazzaroni**, C. Oh
*Applications in Electronics Pervading Industry, Environment and Society* Lecture Notes in Electrical Engineering
pp. 123–130 Conference

**Deep Learning-Based Real-Time Driver Cognitive Distraction Detection**
M. Fresta, F. Bellotti, I. Bochenko, **L. Lazzaroni**, G. Merlhiot, F. Tango, R. Berta
*IEEE Access* vol. 13, pp. 26589–26607 Journal

**Comparing Detection-Based and Motion-Based Preprocessing of Video Frames for Automated Driving Scenario Detection**
M. Cossu, R. Berta, M.A. Bagheri Orumi, **L. Lazzaroni**, J.L. Sauvaget, H. Touil, F. Bellotti
*Applications in Electronics Pervading Industry, Environment and Society* Lecture Notes in Electrical Engineering
pp. 250–257 Conference

**Binary Neural Networks for Real-Time Structural Health Monitoring on Edge Devices**
A. Dabbous, **L. Lazzaroni**, F. Sakr, R. Berta, M. Fresta, F. Bellotti
*2025 IEEE International Conference on Consumer Electronics (ICCE)*
Digest of Technical Papers IEEE International Conference on Consumer Electronics Conference

**Benchmarking Microcontrollers with Ultra-Low Resolution Images Classification**
A. Dabbous, R. Berta, **L. Lazzaroni**, A. Pighetti, F. Bellotti
*Applications in Electronics Pervading Industry, Environment and Society* Lecture Notes in Electrical Engineering
pp. 234–241 Conference

2024

**An embedded end-to-end voice assistant**
**L. Lazzaroni**, F. Bellotti, R. Berta
*Engineering Applications of Artificial Intelligence* vol. 136, art. 108998 Journal

**Adversarial Policy Generation in Automated Parking**
A. Pighetti, F. Bellotti, R. Berta, A. Cavallaro, **L. Lazzaroni**, C. Oh
*ECCV 2024 Workshop on Multimodal Perception and Comprehension of Corner Cases in Autonomous Driving* Workshop

**CBin-NN: An Inference Engine for Binarized Neural Networks**
F. Sakr, R. Berta, J. Doyle, A. Capello, A. Dabbous, **L. Lazzaroni**, F. Bellotti
*Electronics* vol. 13, no. 9, art. 1624 Journal

**End-to-End Dataset Collection System for Sport Activities**
M. Fresta, F. Bellotti, A. Capello, A. Dabbous, **L. Lazzaroni**, F. Ansovini, R. Berta
*Electronics* vol. 13, no. 7, art. 1286 Journal

**YoloP-Based Pre-processing for Driving Scenario Detection**
M. Cossu, R. Berta, L. Forneris, M. Fresta, **L. Lazzaroni**, J.L. Sauvaget, F. Bellotti
*Applications in Electronics Pervading Industry, Environment and Society* Lecture Notes in Electrical Engineering
pp. 418–423 Conference

**Low-Cost, Edge-Cloud, End-to-End System Architecture for Human Activity Data Collection**
M. Fresta, A. Dabbous, F. Bellotti, A. Capello, **L. Lazzaroni**, A. Pighetti, R. Berta
*Applications in Electronics Pervading Industry, Environment and Society* Lecture Notes in Electrical Engineering
pp. 444–449 Conference

**Investigating Adversarial Policy Learning for Robust Agents in Automated Driving Highway Simulations**
A. Pighetti, F. Bellotti, C. Oh, **L. Lazzaroni**, L. Forneris, M. Fresta, R. Berta
*Applications in Electronics Pervading Industry, Environment and Society* Lecture Notes in Electrical Engineering
pp. 124–129 Conference

**Development of Deep-Learning-Based Autonomous Agents for Low-Speed Maneuvering in Unity**
R. Berta, **L. Lazzaroni**, A. Capello, M. Cossu, L. Forneris, A. Pighetti, F. Bellotti
*Journal of Intelligent and Connected Vehicles* vol. 7, no. 3, pp. 229–244 Journal

**Developing a TinyML Image Classifier in an Hour**
R. Berta, A. Dabbous, **L. Lazzaroni**, D.P. Pau, F. Bellotti
*IEEE Open Journal of the Industrial Electronics Society* vol. 5, pp. 946–960 Journal

**Building a Pipeline for Efficient Production of Synthetic Datasets for Improving RL in Automated Driving**
**L. Lazzaroni**, A. Pighetti, F. Bellotti, R. Berta
*Proceedings of SIE 2023* Lecture Notes in Electrical Engineering pp. 351–356 Conference

**Bringing Intelligence to the Edge for Structural Health Monitoring: The Case Study of the Z24 Bridge**
A. Dabbous, R. Berta, M. Fresta, H. Ballout, **L. Lazzaroni**, F. Bellotti
*IEEE Open Journal of the Industrial Electronics Society* vol. 5, pp. 781–794 Journal

**Automated Parking in CARLA: A Deep Reinforcement Learning-Based Approach**
**L. Lazzaroni**, A. Pighetti, F. Bellotti, A. Capello, M. Cossu, R. Berta
*Applications in Electronics Pervading Industry, Environment and Society* Lecture Notes in Electrical Engineering
pp. 352–357 Conference

**Application of a Serious Game for Emotion Elicitation Under Socio-Economic and Trust Based Decision-Making Scenarios for Autistic Adolescents**
F. Ahmed, R. Berta, F. Bellotti, F. Floris, **L. Lazzaroni**, G. Barresi, J.R. Carrion
*Games and Learning Alliance*
Lecture Notes in Computer Science Including Subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics
pp. 170–180 Conference

2023

**Implementing Deep Reinforcement Learning (DRL)-based Driving Styles for Non-Player Vehicles**
L. Forneris, A. Pighetti, **L. Lazzaroni**, F. Bellotti, A. Capello, M. Cossu, R. Berta
*International Journal of Serious Games* vol. 10, no. 4, pp. 153–170 Journal

**Socio-Economic Decision Making and Emotion Elicitation with a Serious Game in the Wild**
F. Ahmed, R. Berta, F. Bellotti, **L. Lazzaroni**, F. Floris, G. Barresi, J.R. Carrion
*Applied Sciences* vol. 13, no. 11, art. 6432 Journal

**Supporting a .csv-based Workflow in MongoDB for Data Analysts**
M. Fresta, A. Capello, F. Bellotti, **L. Lazzaroni**, M. Cossu, R. Berta
*2023 IEEE 32nd International Symposium on Industrial Electronics (ISIE)*
IEEE International Symposium on Industrial Electronics Conference

**Investigating High-Level Decision Making for Automated Driving**
A. Capello, L. Forneris, A. Pighetti, F. Bellotti, **L. Lazzaroni**, M. Cossu, A. De Gloria, R. Berta
*Applications in Electronics Pervading Industry, Environment and Society* Lecture Notes in Electrical Engineering
pp. 307–311 `Conference`

**Explaining a Deep Reinforcement Learning (DRL)-Based Automated Driving Agent in Highway Simulations**
F. Bellotti, **L. Lazzaroni**, A. Capello, M. Cossu, A. De Gloria, R. Berta
*IEEE Access* vol. 11, pp. 28522–28550 `Journal`

**Efficient Uploading of.Csv Datasets into a Non-Relational Database Management System**
M. Fresta, F. Bellotti, A. Capello, M. Cossu, **L. Lazzaroni**, A. De Gloria, R. Berta
*Applications in Electronics Pervading Industry, Environment and Society* Lecture Notes in Electrical Engineering
pp. 9–15 `Conference`

**Developing a Toolchain for Synthetic Driving Scenario Datasets**
M. Cossu, R. Berta, A. Capello, A. De Gloria, **L. Lazzaroni**, F. Bellotti
*Applications in Electronics Pervading Industry, Environment and Society* Lecture Notes in Electrical Engineering
pp. 222–228 `Conference`

**Designing an Interpretability Analysis Framework for Deep Reinforcement Learning (DRL) Agents in Highway Automated Driving Simulation**
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M. Cossu, R. Berta, A. Capello, A. De Gloria, **L. Lazzaroni**, F. Bellotti
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A. Capello, L. Forneris, A. Pighetti, F. Bellotti, **L. Lazzaroni**, M. Cossu, A. De Gloria, R. Berta
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**Explaining a Deep Reinforcement Learning (DRL)-Based Automated Driving Agent in Highway Simulations**
F. Bellotti, R. Berta, A. De Gloria, A. Capello, M. Cossu, **L. Lazzaroni**
*53rd Annual Meeting of the Italian Society of Electronics (SIE), Pizzo (VV)* `National`

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**L. Lazzaroni**, F. Bellotti, A. De Gloria, R. Berta
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**Assessing Versatility of a Generic End-to-End Platform for IoT Ecosystem Applications**
R. Berta, F. Bellotti, A. De Gloria, **L. Lazzaroni**
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M. Cossu, J.L.Q. Villon, F. Bellotti, A. Capello, A. De Gloria, **L. Lazzaroni**, R. Berta
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**L. Lazzaroni**, F. Bellotti, A. De Gloria, R. Berta
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R. Berta, A. Mazzara, F. Bellotti, A. De Gloria, **L. Lazzaroni**
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G. Campodonico, F. Bellotti, R. Berta, A. Capello, M. Cossu, A. De Gloria, **L. Lazzaroni**, T. Taccioli, F. Davio
*Games and Learning Alliance*
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**L. Lazzaroni**, A. Mazzara, F. Bellotti, A. De Gloria, R. Berta
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